

SOFIA PETRUSENKO

Cascais, Portugal | +44 773 771 2377 | sofia.petrusenko988@gmail.com
sofiapetrusenko.dev | github.com/sofiapetrusenko | linkedin.com/in/sofiapetrusenko

PROFESSIONAL SUMMARY

Biomedical scientist and self-taught software engineer working at the intersection of ageing biology and applied AI. Research training across the UK, Portugal, Germany and Ukraine in autophagy, proteostasis, neurodegeneration and longevity, supported by hands-on molecular biology, microscopy and R/Python data analysis. Since 2025 has independently designed, built and operated production systems end-to-end — LLM orchestration pipelines, retrieval-based memory architectures, distributed task processing and cloud deployment — alongside data-driven paid acquisition at scale. Seeking roles where scientific domain knowledge and technical execution meet: longevity and biotech ventures, computational biology, scientific software, or research support.

CORE SKILLS

Software Engineering & AI: Python, FastAPI, PostgreSQL, Redis / RQ task queues, Next.js, REST API integration, LLM orchestration and prompt engineering, retrieval-augmented generation (RAG) and long-term memory design, TTS and media processing pipelines, Docker, Git, DigitalOcean deployment

Data & Bioinformatics: R, RStudio, Python (pandas, NumPy), statistical analysis and data visualisation, BLAST, UniProt, ClustalX, Jalview, Image Lab, CFX Maestro

Molecular & Cell Biology: qPCR, Western blotting, ELISA, immunoprecipitation, enzymatic assays, DNA and protein extraction, mammalian cell culture, bacterial culturing, DNA ligation, restriction digestion, bacterial conjugation, miniprep

Microscopy & Analytical Methods: confocal microscopy, light microscopy, electron microscopy, spectrophotometric analysis, thin layer chromatography, animal dissection

Growth & Performance Marketing: Google Ads and Meta Ads campaign architecture, audience segmentation, creative testing, conversion tracking and attribution, budget allocation, funnel and cohort analytics

Languages: English (fluent), Ukrainian (native), Russian (native)

Other Software: Microsoft Office, Capture One, Adobe Photoshop, Adobe Premiere Pro, DaVinci Resolve

TECHNOLOGY & COMMERCIAL EXPERIENCE

Founder & Engineer — AI Content Automation Platform | Independent venture

Cascais, Portugal (remote) | 2025 – present

- Designed and built a full-stack platform for automated media content production and distribution: LLM-driven generation pipeline, text-to-speech synthesis and video assembly, publishing via the YouTube Data API, analytics ingestion and monitoring.
- Architected a seven-stage processing pipeline (script → voiceover → asset sourcing → assembly → subtitling → metadata → thumbnail) orchestrated through asynchronous background workers.
- Stack: Python / FastAPI, PostgreSQL, Redis with RQ workers, Next.js front end, deployed on DigitalOcean; integrated third-party APIs for LLM inference, speech synthesis and stock media.
- Operates a multi-channel content network with human-in-the-loop editorial control, including a longevity and healthspan channel drawing directly on biomedical subject-matter expertise.
- Sanitized platform code (architecture, pipeline, tests) is open-sourced at github.com/sofiapetrusenko/media-automation-platform.
- Extended the system to multi-language output, launching Spanish-language production from the same pipeline.

Paid Acquisition & Performance Marketing Specialist | Independent / contract

Cascais, Portugal | 09/2025 – present

- Plan, launch and optimise paid acquisition campaigns on Google Ads and Meta, managing budgets against defined cost-per-acquisition and return targets.
- Build and test creative and landing-page variants, running structured experiments and interpreting results statistically rather than anecdotally.
- Own tracking, attribution and reporting across the funnel; use performance data to reallocate spend and iterate on targeting.

AI Engineer — Conversational Agent with Persistent Long-Term Memory | Independent project

09/2025 – 12/2025

- Designed and implemented a therapeutic-style conversational agent built on an Obsidian Markdown knowledge base, giving the system durable memory across sessions rather than single-conversation context.
- Developed the memory layer end-to-end: structured note schema, automated extraction and writing of salient information, and retrieval of relevant prior context at inference time.
- Engineered prompting and conversation-state handling to maintain continuity, coherence and appropriate tone over extended multi-session interactions.

RESEARCH EXPERIENCE

Research Intern | Dr. Adrien Rousseau, MRC PPU, School of Life Sciences

Dundee, Scotland, UK | 09/2024 – 01/2025

- Investigated the role of stress-induced protein phosphorylation in *Saccharomyces cerevisiae* and its relationship to longevity.
- Used R and Python to analyse experimental datasets and identify patterns and correlations linked to yeast lifespan.
- Collaborated on experimental design, protocol development and interpretation of findings; presented results in team meetings.
- Contributed to drafting a manuscript on the implications of the results for ageing mechanisms.

Research Intern | Sandra Tenreiro & Paula Macedo Labs, NOVA Medical School

Lisbon, Portugal | 06/2023 – 08/2023

- Studied molecular mechanisms of ageing relevant to neurodegeneration and retinal disease, with a focus on proteostasis.
- Worked with cellular models ranging from yeast to human cells, as well as brain and retinal organoids.
- Supported research into therapeutic approaches for Parkinson's disease, diabetic retinopathy and age-related macular degeneration, in collaboration with clinicians.

Research Technician | Dr. James Murray Lab, Institute of Life Sciences

UK | 05/2022 – 09/2022

- Investigated autophagy signalling and its alteration during ageing and in disease contexts including cancer, neurodegeneration and diabetes.
- Contributed to research on autophagy in immunity and pathogenic infection, including SARS and SARS-CoV-2-related work.

Research Intern | Prof. Dr. Iain D. Couzin, Max Planck Institute of Animal Behaviour

Konstanz, Germany | 03/2022 – 05/2022

- Researched collective animal behaviour and the role of social influence across biological systems.
- Worked across experimental and theoretical approaches to explore group dynamics and coordinated action from cells to animals and human societies.

Laboratory Assistant | Anatoliy I. Dragan Lab

Kyiv, Ukraine | 02/2021 – 09/2021

- Supported an experimental study of DNA hydration using Pressure Perturbation Scanning Microcalorimetry.

EDUCATION

BSc (Hons) Biomedical Sciences | University of Dundee

Scotland, UK | 09/2022 – 05/2025

- Placement and project work focused on ageing biology, autophagy, proteostasis and longevity mechanisms.

BSc Biology | Taras Shevchenko National University of Kyiv, Institute of High Technologies

Kyiv, Ukraine | 09/2020 – 04/2022

ADDITIONAL

- Training: ICH E6 Good Clinical Practice (GCP) Investigator Site Training.
- International research exposure across the UK, Portugal, Germany and Ukraine.
- Comfortable working autonomously and shipping projects end-to-end, from specification through build to live operation.